21/12/24 M/S

USN | | | | | | | | | | [22CSE369]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

STORAGE MANAGEMENT AND DATA CENTER

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5 x2=10**
1.	**a.**	Identify the main components of a disk drive.	L1	CO1	**02**
	b.	What does ILM stand for?	L1	CO1	**02**
	c.	Describe the role of a storage controller in a storage system.	L2	CO4	**02**
	d.	Describe the concept of redundancy in business continuity.	L2	CO6	**02**
	e.	What are the characteristics of an outstanding data center design?	L1	CO3	**02**
		PART - B			**6x5=30**
2.	**a.**	Define data center. Explain why data center is essential for the organizations now a days.	L2	CO1	**05**
		OR			
	b.	Explain the significance of data availability in data centers.	L2	CO1	**05**
3.	**a.**	Describe the role of Information Lifecycle Management (ILM).	L2	CO1	**05**
		OR			
	b.	Discuss the importance of redundancy in business continuity.	L2	CO5	**05**
4.	**a.**	Illustrate the differences between Direct Attached Storage (DAS) and Network Attached Storage (NAS).	L3	CO3	**05**
		OR			
	b.	Show how disk performance factors can affect storage systems.	L3	CO2	**05**
5.	**a.**	List the characteristics of an outstanding data center design.	L2	CO3	**05**
		OR			
	b.	Explain the function of the Access Layer in a data center.	L2	CO4	**05**
6.	**a.**	Discuss how disaster recovery plans can benefit a data center.	L2	CO6	**05**
		OR			
	b.	Explain the concept of logical block addressing.	L2	CO4	**05**
7.	**a.**	Demonstrate the importance of cable management in data centers.	L3	CO4	**05**
		OR			
	b.	List and describe the different types of data storage technologies.	L1	CO2	**05**
		PART - C			**6x10=60**
8.	**a.**	Discuss the significance of data availability in modern business environments.	L2	CO1	**10**
		OR			

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	b.	Illustrate the importance of redundancy in business continuity planning.	L3	CO6	**10**
9.	**a.**	Describe how disk drive performance factors can influence overall storage system performance.	L2	CO2	**10**
		OR			
	b.	Compare the scalability and performance between DAS, NAS, and SAN in enterprise environment.	L2	CO2	**10**
10.	**a.**	Discuss how virtualization technologies affect data center architecture.	L2	CO4	**10**
		OR			
	b.	Discuss the implications of emerging technologies on data center management.	L2	CO1	**10**
11.	**a.**	Discuss the impact of security measures on data center operations.	L2	CO5	**10**
		OR			
	b.	Explain the significance of information availability in a data center.	L2	CO5	**10**
12.	**a.**	Explain the role of data center transport layers in network communications.	L2	CO4	**10**
		OR			
	b.	Describe how the evolution of storage technology impacts data center architecture.	L2	CO3	**10**
13.	**a.**	Evaluate the role of Fibre Channel (FC) connectivity and topologies in the performance and reliability of a SAN. How does FC enhance the functionality of SAN environments?	L4	CO4	**10**
		OR			
	b.	Examine the factors that influence disk drive performance, including disk service time and the fundamental laws governing disk performance.	L3	CO2	**10**